

Security Clearance: Secret — held through August 2025

EDUCATION**Cornell University**, College of Engineering, Ithaca, NY
Mechanical Engineering, **GPA: 4.050****Expected May 2027**

Selected Coursework: Engineering Simulations using ANSYS • Propulsion of Aircraft and Rockets • Mechatronics • Spaceflight Mechanics • Heat Transfer • Mechanics of Engineering Materials • Statics • Thermodynamics • Dynamics • System Dynamics • Fluid Mechanics

SPECIALIZED SKILLS

CAD & Analysis: ANSYS Workbench, SolidWorks, Autodesk Inventor, Creo Parametric, CAM

Software: MATLAB, working knowledge of Java and Python, Excel

Manufacturing: Engineering Drawings, Basic GD&T, Composite layups, Machining using manual mills and lathes

ENGINEERING EXPERIENCE**Monocoque Subteam Lead**, *Cornell FSAE Racing*, Cornell University**Fall 2023-Present**

- Collaborate with 60+ students to design, build, and race an electric Formula SAE racecar
- Responsible for the design, manufacturing, and analysis of carbon fiber monocoque including ply schedule, geometry, and in-house composite layups
- Coordinate system packaging and integration with all 5 mechanical and 3 electrical subteams
- Designed and optimized the protective structures including the roll hoops and front impact structures
- Presented and defended engineering decisions throughout multiple stages of design review and feedback

Mechanical Engineering Intern, *Quantum Space*, Rockville, MD**May 2026-Aug. 2026**

- Designed dynamic mass simulator for spacecraft propellant tank structures vibration qualification testing
- Verified structural integrity through finite element analysis, validating load paths and margins of safety
- Designed CAD models and engineering drawings, applying GD&T and tolerance stack-up analysis to create a bearing-contact shear interface
- Researched and presented findings on fabrication methods for the production of deployable composite booms

Mechanical Fabrication Intern, *Johns Hopkins University Applied Physics Lab*, Laurel, MD**May 2025-Aug. 2025**

- Manufactured advanced prototype and flight hardware under the Materials and Processing Engineering Group
- Supported composite fabrication processes, including layups, tooling preparation, and bonding assemblies
- Created 2D ply designs, prepared additively-manufactured tooling, and laid up composite prototype nosecone supporting NASA's Dragonfly space mission

Manufacturing Engineering Intern, *Johns Hopkins University Applied Physics Lab*, Laurel, MD**2021-2023**

- Summer and academic year-long internship program under the lead manufacturing engineer, from 6/2021 - 5/2023
- Assisted composite and inspection engineers, created detailed CAD models to optimize space in machine shop
- Analyzed and created infographics of raw historical manufacturing data from machine shop using Excel, presented models to project leaders

LEADERSHIP AND ACTIVITIES**Breakdancing Club President**, *Cornell's Absolute Zero Breakdance Club*, Ithaca, NY**May 2025 - May 2026**

- Delegated tasks, managed group dynamics, directed communications with other organizations, planned practice
- Engaged with local breaking community through hosting beginner-friendly workshops and open practices

Founder of Region 7 Care Package Program, *Region 7 Acro United*, Laurel, MD**2022-2023**

- Worked with R7 Acro United, a 501C3 non-profit organization, to develop a care package program for injured acrobatic gymnasts in Region 7 (Maine to South Carolina)
- Provided seed money, planned fundraisers with the executive board, ran fundraisers at competitions